

The Utility Problem

Electromobility, as one of the leading innovative trends of today and also one of the main instruments to achieve the goals of the Paris Agreement, is very important for utilities in terms of load growth on the grid (due to the rapid rise in numbers of EV chargers). This trend is already at the forefront and will continue to grow in importance.

One of the tools which utilities use for keeping the load at individual consumption points “under control” is assigning fixed reserved capacity for each consumer - usually by master circuit breaker (MCB). The maximal reserved capacity of the consumption point describes the maximum power that can be consumed at a given point. Based on the utility billing scheme, the customer might:

- A. Pay a specified fixed monthly amount for this capacity, and/or
- B. For situations when this capacity is exceeded, either a fee is charged, or the main circuit breaker is tripped.

These capacity fees and/or peak load charges might be quite significant, especially for industrial customers, so it is important to optimize them. Utilities also use MCB (and maximal demand) values for network dimensioning, assessing the grid's capacity.

To optimize these values, utilities or EV charger administrators usually focus only on the maximum load of EV chargers and the maximum load of other electricity-consuming equipment. However, this does not describe the real world, in which maximum load on all elements only occurs in rare cases. Thus, there is a need to look at this issue in more detail using synchronized measured data. This methodology should work based on probabilities and should more closely approximate real behaviour. Just synchronizing the data from the meters and EV chargers can be a challenging task as the data from the different systems will likely be in different formats. For such analysis, the data cleaning and possible data correction might also need to be solved.

This type of analysis can identify under-utilized, (unused) reserved capacity at a selected location (consumption point), which can save significant amounts of capacity fee payment for consumers. Consumers might also be able to simulate how many EV chargers can be added before the maximal capacity is exceeded. On the utility side, this can save investment funds, as it will not be necessary to upgrade the grid with, for example, a higher-diameter conductor or a new station with a higher capacity. This also saves resources and, therefore, CO₂.

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have several datasets ready. These datasets are as follows:

1. Consumption time-series from the utility meter (consumption point with EV charger).
2. Consumption time-series or consumption data (one value per changing session) from EV charger(s).

3. Master circuit breaker value or maximum capacity at location (consumption point) and maximum load of EV chargers.
4. Typical EV charger time-series - for simulation purposes.

The presence of these datasets alone does not guarantee certain success. Once the data has been collected, it is necessary to move on to synchronizing the meter and EV charger data, which as mentioned previously, can be a challenging task. However, with the Awesense Energy Transition Platform, everything is already in place and analysts don't have to deal with this at all. An analyst can determine if the capacity at a location (consumption point) is already at the edge of the contracted maximum capacity or if and how many new EV chargers can be installed at that location (consumption point). These new EV chargers are added to the selected location (consumption point) until the contracted capacity is reached. It is possible to work with the probability of all devices running simultaneously and to calculate the maximum capacity, for example, 99% of the highest load values can be used and extreme values can be discarded (worst case situations that occur only 1% of the time).

It is also possible to find out what is the maximum number of EV chargers that have ever been in use or how often have a particular number of EV chargers been in simultaneous use. What can be determined as well is the EV contribution to the total consumption point load over time, especially during peak hours or what percentage of a consumption point's total consumption was due to EV charging.

To get the necessary datasets for the analysis of free capacity in places with EV chargers, the Awesense Energy Data Model (EDM) interface offers multiple methods:

1. Consumption time-series from the utility meter (consumption point):
Querying time series values is very simple in EDM. It even offers functions to aggregate this data (into hourly data, for example), as often 15-minute data is not needed for analysis.
2. Consumption time-series or consumption data (one value per charging session) from EV charger (or chargers).
The data is obtained through the same process as the previous dataset or time-series data is created from a single measured value and the duration of the charging time.
3. Master circuit breaker value or maximum capacity at location (consumption point) and maximum load of EV chargers.
Information about the capacity of the location (consumption point) and the maximum capacity of the EV chargers is included in the EDM model and is therefore very easy to get.
4. Typical EV charger time-series - for simulation purposes.
To get this data, it is necessary to filter for EV charger elements and then produce typical charging time series curves.

Finishing this analysis is just a question of building the algorithms to produce summed kW for all EV chargers and meters at the location (consumption point) and comparing it to the Master

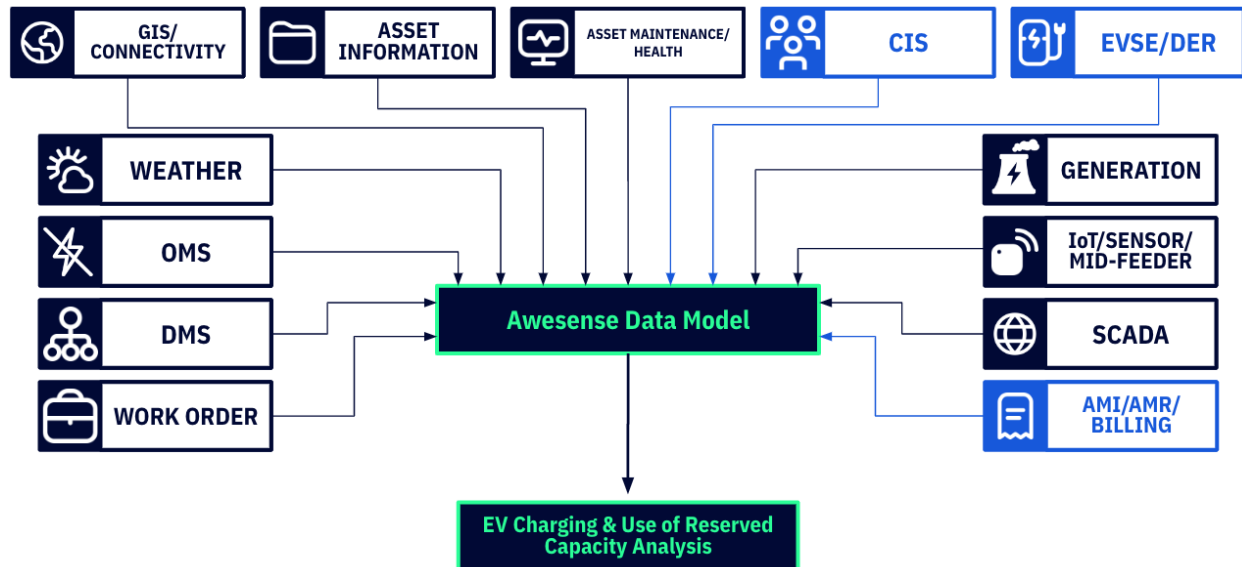
EV Charging & Use of Reserved Capacity Analysis

[UC09-01] - Awesense Energy Transition Platform Use Case



circuit breaker (MCB) value to identify available capacity. Adding new EV chargers to location (consumption point) will provide information on how many new EV chargers can be installed.

Required Data Sets



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.